

Answers — items 1 and 2

The key to the two items we did **in the room**. Items 3 and 4 are **PS4**; their solutions post to bCourses after the deadline, not here.

1 • Which curve is the hyperbola

The v -nullcline, $v = \alpha_2 / (1 + u)$, is the hyperbola: with $\gamma = 1$ there is no inflection because $1 / (1 + u)$ has none — it is the session-4 curve with $n = 1$, the one that could never be steep. On log axes it is the nearly straight line of slope -1 at large u . The u -nullcline, $u = \alpha_1 / (1 + v^{2.5})$, is the sigmoid, and its steep drop near $v \approx 1$ is what lets the two curves meet three times.

Worth noticing: the right-hand panel is the same α_1, α_2 with β also set to 1. No promoter strength rescues it — the bound $g_1 g_2 < 1$ holds everywhere. That is the sentence from the lecture, *no cooperativity, no switch*, in its correct form: **no cooperativity on either arm**.

2 · pTAK117's three crossings

At B, $u = 1.317$, $v = 6.734$:

$$g_1 = 2.5 \cdot \frac{1.317 \cdot 6.734^{1.5}}{1 + 6.734^{2.5}} = 0.485, \quad g_2 = 1 \cdot \frac{6.734}{1 + 1.317} = 2.907, \quad g_1 g_2 = 1.41.$$

$\lambda = -1 \pm \sqrt{1.41} = -1 \pm 1.19$, so $\lambda = +0.19$ and -2.19 . A saddle: one direction leaves, slowly. (jacobian from posb gives the same four numbers by finite differences — check yourself.)

For the record, A gives $g_1 g_2 = 0.62$ and C gives **0.008**; both stable.

► **Why the product decides.** A perturbation has to go round the loop: a change in u alters dv/dt by g_2 , the resulting change in v alters du/dt by g_1 , and what comes back is the product. A gain of 12 on one arm and 0.0006 on the other is a loop that returns 0.008 of what left — deeply stable. Neither factor alone means anything; it is the same reason a two-stage amplifier's gain is a product.

At C, $u = 156$ and $v = 0.1$: LacI is high and cl is essentially off. LacI is in charge — the u -nullcline is vertical there, and $g_2 \approx 0.001$ says cl has no purchase on it at all ($g_2 = 6 \times 10^{-4}$).

► **Which is the low state.** The reporter (GFP) sits with cl, so **C, the LacI-high state, is the low state**, and A is the high state. Its loop gain is **0.008** against **0.62** for A: C sits far from marginal, A sits close to it, which is the paper's *the low state is more stable than the high state*. The asymmetry in the α 's (156 against 15.6) is the asymmetry in the gains. (The bimodal culture in Fig. 5c is a different fact — a population straddling the fold at which the *low state* is annihilated — and it is Thursday's.)